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RESEARCH ARTICLE

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Gas/particle flow and pressure variation characteristics of Ende pulverized coal gasifier equipped with ejector

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Abstract

Ende pulverized coal gasifier (EPCG), as a typical representative of circulating fluidized bed gasifier, has been widely used in gasification field. The return conduit of EPCG is generally not equipped with a feedback control device (FCD), which tends to result in a low solid circulation rate (G_s) and high unburned combustibles in fly ash. To address these issues, a new technique using the ejector as the FCD was designed to provide a stable pressure barrier and improve solid circulation. Preliminary thermal experiments demonstrated the feasibility of improving the EPCG with injector system. Currently, the specific mechanisms that contribute to further optimize this technology have not been systematically studied. Considering the large impact of ejector operation on solid circulation rate and stability as well as syngas production, this work adopts a gas/particle 1:15 cold EPCG bench to investigate the gas/particle flow and pressure variation characteristics before and after ejector operation and under different working medium pressures (P_p). When the ejector operates at the design pressure $P_s = 0.187$ MPa, the outlet pressure of the return conduit is slightly larger than the furnace bottom pressure, thus preventing the reverse gas flow at the furnace bottom. The particle velocity and concentration distribution in the furnace are basically consistent with that without the ejector, showing a similar central spout region and near-wall annulus flow region. As P_p increases from 0.04 to 0.24 MPa, the average particle velocity in the return conduit increases from 0.95 to 2.4 m/s, the particle concentration decreases from 0.088 to 0.051, and the solid circulation rate increases from 0.37 to 0.53 kg/h. The ejector outlet pressure gradually increases, the inlet pressure decreases, and the pressure barrier provided by the ejector rises from 1052 to 3640 Pa. When P_p is 0.12 MPa, the ratio of G_{N2}/G_{air} is 0.91%, which can achieve the purpose of continuous and stable pressurization with a small amount of nitrogen.

KEYWORDS

ejector, Ende pulverized coal gasifier, particle concentration, pressure variation, solid circulation

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1 | INTRODUCTION

Atmospheric circulating fluidized bed gasifier (AT-CFBG) is a typical gasification technology for converting coal into more easily purified fuel gas, such as industrial fuel gas, city gas, and chemical synthesis gas, with the advantages of high gasification intensity, simple operation, and so on.[1,2] Since the pressure at the bottom of AT-CFBG is greater than the pressure at the return conduit outlet, a feedback control device (FCD) is commonly installed to prevent recirculation at the furnace bottom, which plays a key role in stabilizing the solid circulation process and increasing the solid circulation rate (G_s). For FCD, several studies had examined the effect of FCD structure and operating parameters on particle flow characteristics, solid circulation rate, and operational stability.[3–7]

Ende pulverized coal gasifier (EPCG), as a kind of AT-CFBG, has been widely used in gasification field. However, the return conduit of EPCG is generally not equipped with FCD. When the solid inventory in the return conduit reaches a certain height, it can overcome the resistance of the reverse gas flow and re-enter the furnace through gravity self-balancing. If the solid inventory is too low to form a pressure barrier, part of the flue gas from the furnace will pass through the return conduit and reduce the cyclone efficiency.[8–10] In this process, the furnace pressure and coal feeding rate will affect the solid circulation effect, and the flue gas flow can easily flow back to the cyclone separator through the return conduit, thereby reducing the solid circulation rate and even causing serious mass and pressure fluctuations. The above issues increase the unburned combustible of fly ash and fly ash discharge, reducing the economic benefits of EPCG and the effective syngas concentration. To solve these problems, Liu et al. developed a novel technique. In this technology, the ejector,[11–13] which is mainly used in refrigeration, desalination, refrigeration, and thermal power plants, is used as FCD to provide a stable pressure barrier with a small amount of working medium and is suitable for achieving continuous solid flow. Currently, only a few studies have reported its use for capturing solid particles.[14] Among them, Gül applied a gas–solid ejector to the return conduit of a cold CFB model and used N_2 as the ejector motive gas. Experimental results showed that the ejector could provide the required pressure barrier for the CFB loop.[15]

To verify the effectiveness of the ejector system, an ejector was installed on the return conduit of a 45 000-Nm^3/h EPCG. The operating results showed that, compared with the original EPCG, the ejector could provide a stable pressure barrier and effectively prevent gas backflow under a steam pressure of 0.5 MPa and a mass flow rate of 617 kg/h. The effective syngas concentration increased from 73.3% to 74.5%, and the unburned combustible content in fly ash decreased from 26.9% to 19.9%.[1] Furthermore, a cold modeling test was used to measure particle concentrations and velocities under six nozzles arranged asymmetrically and symmetrically at the furnace bottom. The asymmetric arrangement has a stable and higher solid circulation rate (0.37 kg/h) compared with the symmetric arrangement (0.34 kg/h).[16] In addition, ejector nozzle abrasion is also an important issue, which can seriously affect combustion stability and effective syngas concentration. Since the radial velocity near the nozzle zone is as high as 10–12 m/s, the nozzle surface will be significantly eroded during long-term operation, and the Ni-based structure of the original material will be seriously damaged.[17] To solve this problem, the nozzle structure was further optimized to reduce the surrounding radial velocity.

Currently, published investigations on pulverized coal gasifier have involved gas/solid modeling tests,[18–20] industrial scale studies,[21,22] and numerical simulations.[23,24] For the gas/solid modeling tests, relevant research has shown that a comprehensive study of the gas–solid two-phase mechanism can largely reflect the actual flow characteristics during the gasification operation and helps to further optimize equipment operation and gas production rate.[18–20] In this work, considering that air distribution plays a crucial role in the formation of gas–solid flow uniformity and affects the actual mixing and heat exchange, as well as the large impact of ejector operation on G_s, circulating solids stability and syngas production, a 1:15 gas/particle cold bench was established for a 45 000-Nm^3/h EPCG to investigate the gas/particle flow and pressure variation characteristics before and after ejector operation and under different working medium pressures (P_p). The main objectives of this study are to determine (1) whether the ejector can achieve the purpose of preventing backflow; (2) the influence of P_p on G_s and the circulating solid stability; (3) the effect of P_p on the particle flow characteristics in the furnace. The relevant results and conclusions will help to optimize and transform the EPCG design.

2 | METHODOLOGY

2.1 | Research object

Figure 1 shows a schematic diagram of an EPCG equipped with an ejector system. The system mainly consists of a reactor, a cyclone separator, an ejector

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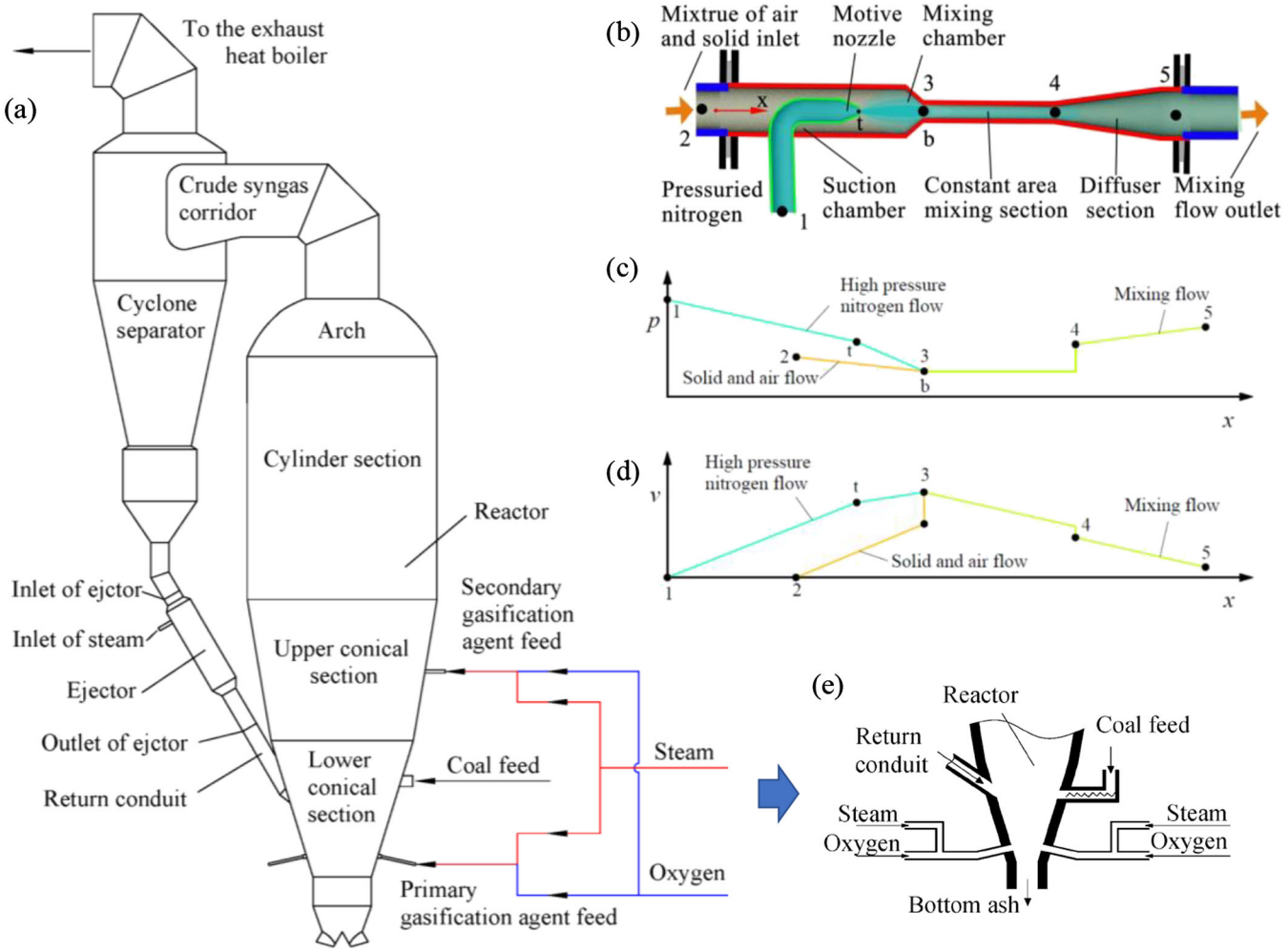


FIGURE 1 Schematic diagram of the EPCG equipped with an ejector system: (a) EPCG equipped with ejector; (b) schematic diagram of steam-solid ejector; (c) pressure profile of ejector; (d) velocity profile of ejector; (e) air distribution at the bottom of the EPCG

system, and a return conduit. The pulverized coal is fed into the bottom of the reactor through the coal feed channel. Primary and secondary gasification agents consisting of oxygen and superheated steam enter the reactor through their respective gasification nozzles. Compared with conventional CFBG,[25] this structural design eliminates the air distribution system at the bottom, which can solve the problems of nozzle erosion and solid backflow into the air chamber of CFBG. Most importantly, the bottom of the cyclone separator is connected to the inlet of the ejector, and the outlet of the ejector is connected to the return conduit at the furnace bottom. The steam used in the ejector comes from the main steam pipeline with a design pressure of 0.5 MPa and a temperature of 300°C.

As shown in Figure 1b, the ejector structure consists of five parts: suction chamber, motive nozzle, mixing chamber, constant-area mixing section, and diffuser section. Figure 1c,d presents the pressure and velocity distributions along the ejector axis, respectively. Inside the ejector, the high-pressure steam (point 1) expands to a back pressure b in the motive nozzle, which is lower than the pressure of the gas and solid mixture (point 2). Therefore, the mixture of gas and solids is entrained by the high-temperature steam through the inlet nozzle to the suction chamber, and the two streams are then mixed together in the mixing chamber, and the pressure is further reduced to the back pressure b (point 3), while the fluid velocity increases. Turbulent mixing occurs in the constant-area mixing section, resulting in an exchange of energy and momentum, and the velocity field tends to decrease from point 3 to point 4. In the diffuser section, the velocity continues to decrease and the pressure increases at the diffuser from point 4 to point 5. A detailed description can be found in reference.[1]

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2.2 | Experimental system

For small-scale modeling tests, it is virtually impossible to replicate all the physical and chemical processes of a full-scale industrial gasifier, but compared with industrial gasifier experiments, results from a great number of different small-scale cold-flow tests showed that reliable predictions could be made from the scaled-down model tests.[26–29] In this modeling test, to ensure that the experimental results are closer to the actual gasifier, some similarity criteria should be met: (1) the model furnace is a 1:15 scale version of a full-scale industrial gasifier, (2) the gas flows are in the second self-modeling region, (3) the ratios of the gas flow momentum values in the model furnace are equal to those in the actual gasifier, and (4) according to the Stokes criterion, pulverized coal in the gasifier should be modeled using solids with an average particle size of 1.2 mm due to their flow similarity.[16]

Figure 2 shows a schematic diagram of the experimental system with a modeling ratio of 1:15. The system consists of six nozzles, a cyclone separator, a furnace model, an ejector, a return conduit, and a particle-velocity optical-fiber probe (PV-6) measurement system. All parts are made of transparent perspex for easy visual observation. The total height of the furnace is 1.3 m, and $D = 313$ mm is the furnace diameter. Due to its high sensitivity, high accuracy, and low signal loss, the PV-6 probe is used to measure particle velocity and concentration. Comprehensive information on the use of PV-6 is reported in references.[30–33] During the experiment, millet was used as the bed material since the coal particles did not meet the steradian and reflectance properties required for PV-6. The mean diameter, real density, and bulk voidage of the millet granules were 1.2 mm, 1500 kg/m^3, and 0.4, respectively. Considering that the actual working medium of the ejector was superheated steam, if superheated steam was used in the two-phase cold experimental system, the steam would condense into supercooled water due to the ambient temperature, thus affecting the particle flow behavior. Therefore, high-pressure nitrogen was used instead of superheated steam. Due to the different thermodynamic properties of nitrogen and superheated steam, the ejector was redesigned to

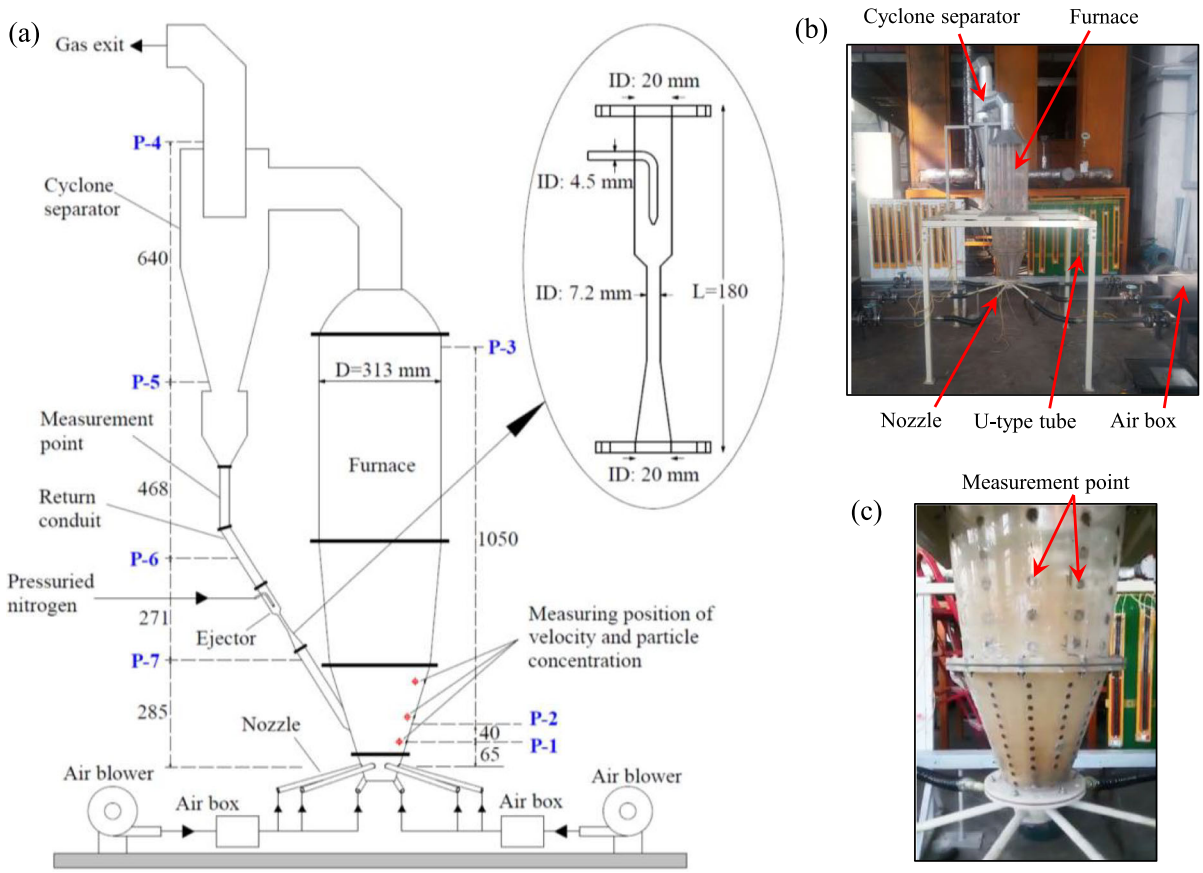


FIGURE 2 Gas/particle experimental measurement system: (a) schematic diagram of the experimental measurement system; (b) photo of experimental system; (c) photo of measuring point layout

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achieve the same performance as the industrial equipment, that is, the ejector outlet was slightly larger, and the design pressure of nitrogen at this time was $P_s = 0.187$ MPa.

2.3 | Experimental method

Before each experiment, the millet particles were evenly spread on the furnace bottom, and the height of the solid inventory was 0.09 m. Then, the transporting air entered the furnace at a speed of 90 m/s from six nozzles, thereby forming a central spouting airflow and a peripheral fluidizing airflow in the furnace, which carried the millet from the furnace bottom to its upper part. The millet then entered the cyclone separator, and a portion of the millet was separated and returned to the furnace. Once the stable operation of the system was achieved, the following aspects were measured:

1. Pressure distribution and variation law

To determine the pressure distribution of the system, seven pressure measuring points marked from P-1 to P-7 are arranged along the solid circulation loop. The specific locations of the pressure measuring points are shown in Figure 2. Three points are arranged along the furnace height, and four points are arranged along the cyclone separator and return conduit.

2. Particle velocity and concentration distribution

The flow field at the intersection of the lower return pipe and the multiple streams of gasification agent is complex, so particle velocity and concentration are measured in this area. Section 1 and Section 2 shown in Figure 3 were chosen to measure particle velocity and concentration distributions at P_s and P_i (0.03, 0.06, 0.12, 0.18, and 0.24 MPa). The origin of the x–y coordinates is on the horizontal plane at the center of the six nozzle outlets, and the radial position is recorded as x/X_0, where $X_0 = D/2$. Three cross-sections of $y/D = 0.2 \sim 0.7$ are established for measuring particle velocity and concentration, and the distance between each two measuring points was 5 mm.

3 | RESULTS AND DISCUSSION

3.1 | Comparison of EPCG with and without ejector

To determine whether it is possible to prevent the reverse airflow and its effect on the particle flow characteristics at the design pressure of $P_s = 0.187$ MPa, the system pressure, particle velocity, and concentration distribution are first investigated without and with ejector operation. Figure 4 shows the pressure profiles at $P_0 = 0$ MPa and $P_s = 0.187$ MPa. When the ejector is operated at $P_s = 0.187$ MPa, the outlet pressure of the return conduit (P-7) is slightly larger than the bottom pressure (P-2), thus preventing reverse airflow. The values of P-1 and P-2 at the furnace bottom are obviously higher than those at $P_0 = 0$ MPa. This is because the mixing airflow from the ejector outlet enters the furnace obliquely at a certain velocity, which partially blocks the upward airflow in the

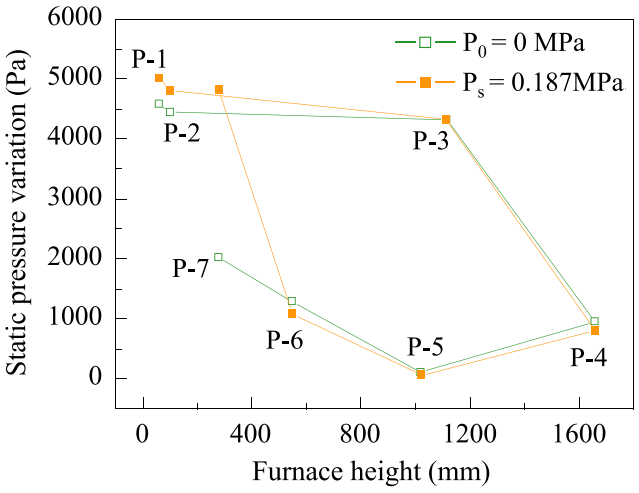


FIGURE 4 Pressure curves at $P_0 = 0$ MPa and $P_s = 0.187$ MPa

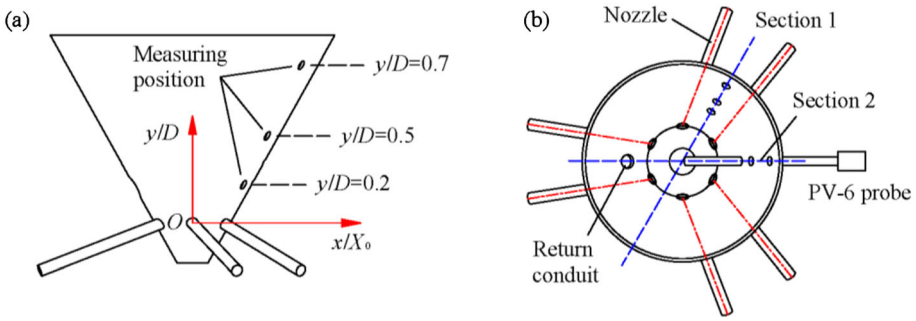


FIGURE 3 Schematic diagram of measuring point layout: (a) main view; (b) top view

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furnace, resulting in a decrease in its speed and an increase in pressure. The pressure value at P-3 is basically the same as that of $P_0 = 0$ MPa, indicating that the pressure at the furnace upper part is not significantly affected by the ejector operation. The values of P-4, P-5, and P-6 are slightly smaller than those at $P_0 = 0$ MPa, which is due to the injection effect of the ejector, which causes the pressure to drop.

Figure 5 shows the particle velocity distribution at P_0 and P_s pressures. As shown in Cross-section 1 in Figure 5a, the particle velocity is positive in the central region of the three sections with $y/D = 0.2$ to 0.7 but negative in the nearby regions. This reflects the distribution characteristics of the central spout region and the near-wall annulus region. This is because the peripheral fluidized airflow has a low momentum that cannot fully lift the particles, causing the particles to recirculate in the near-wall region.[34,35] As the value of x/X_0 increases, it can be seen that the particle velocity under P_s is initially lower than that of P_0, but when x/X_0 increases to a certain value, that is, the x/X_0 values on $y/D = 0.2$, 0.5, and 0.7 are respectively larger than 0.2, 0.17, and 0.1, the particle velocity begins to increase. This is because the return conduit is on the left ($x/X_0 < 0$) and is in the same longitudinal section as Section 1. The high-velocity mixed air/particle flow from the ejector outlet enters the furnace in an oblique direction, which inevitably blocks the upward airflow from the furnace bottom. At the same time, the mixed flow squeezes the central spouting flow to the right region ($x/X_0 > 0$), causing multiple air/particle streams to turn upward in this region, further increasing the particle velocity. When $x/X_0 = 0$, the particle velocities at P_0 pressure at $y/D = 0.2$, 0.5, and 0.7 are 4.82, 5.33, and 3.68 m/s, respectively. Compared with the original values of 6.13, 5.75, and 4.62 m/s, the reductions are 27.12%, 7.88%, and 25.54%, respectively.

Figure 5b shows the particle velocity distribution on Section 2. The particle velocity under P_s is generally lower than that of P_0, indicating that the influence of the ejector operation on the particle velocity on Section 1 is more obvious than that on Section 2. When $x/X_0 = 0$, the particle velocities at P_0 pressure at $y/D = 0.2$, 0.5, and 0.7 are 4.83, 4.12, and 3.82 m/s, respectively. Compared with the original values of 4.91, 4.91, and 4.38 m/s, the decrease is 1.65%, 19.27%, and 14.7%, respectively.

The particle concentration distributions at P_0 and P_s are illustrated in Figure 6. The particle concentrations on Sections 1 and 2 show higher values near the wall

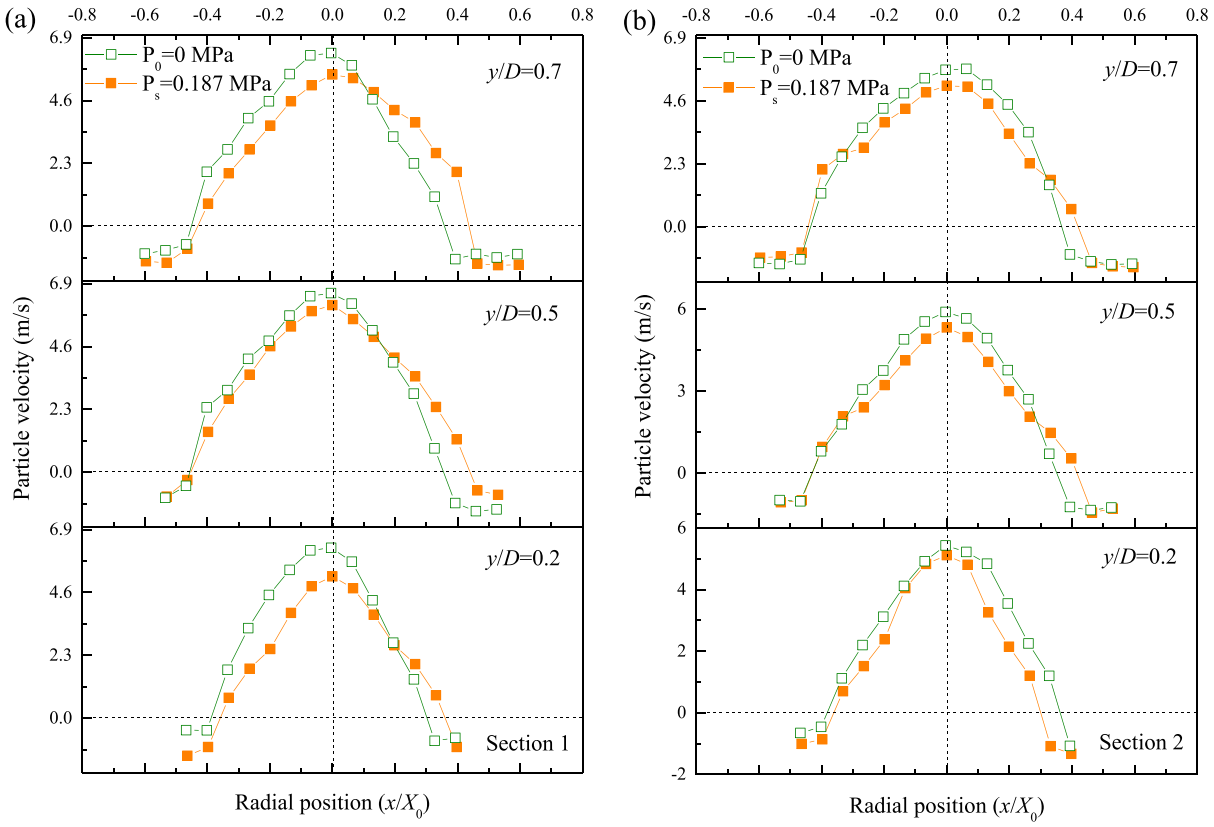


FIGURE 5 Particle velocity distribution at P_0 and P_s pressures: (a) particle velocity curve for Section 1; (b) particle velocity curve for Section 2

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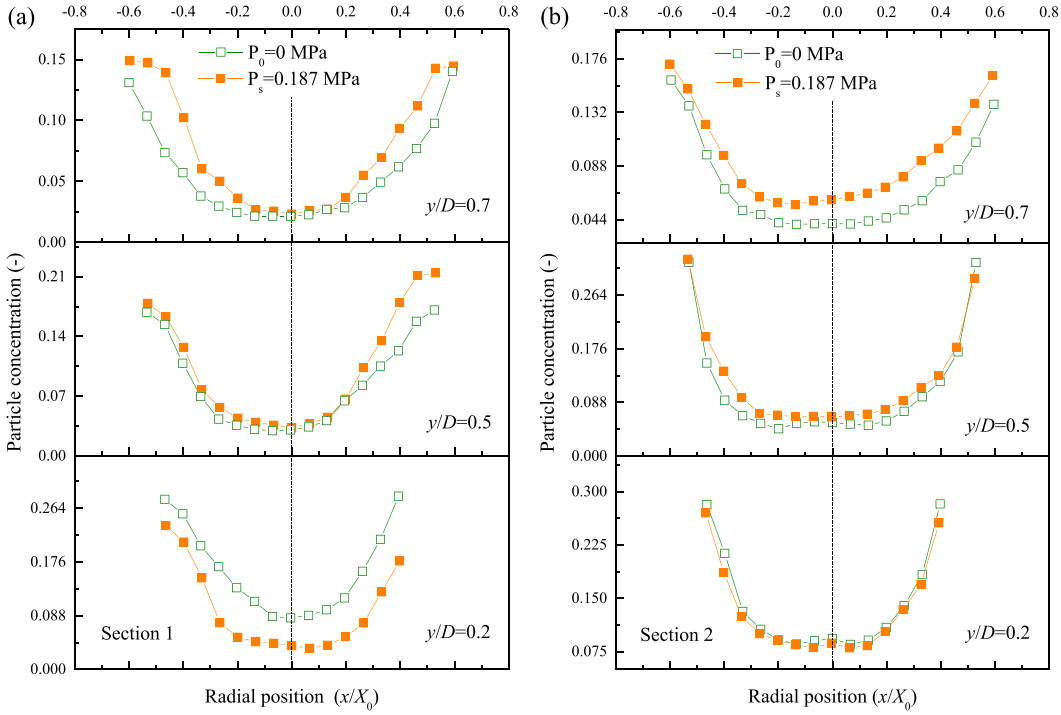


FIGURE 6 Particle concentration distribution at P_0 and P_s pressures: (a) particle concentration curve for Section 1; (b) particle concentration curve for Section 2

and lower values in the central area, which is similar to that reported in the literature.[36] Figure 6a shows the particle concentration distribution on Section 1. It can be seen that the particle concentration under ejector operation (P_s) is higher than that at P_0 except for $y/D = 0.2$. This is because when the ejector is running, there is no solid accumulation in the return conduit, and the reverse return flow entrains more particles into the furnace, resulting in an increase in the number of particles in the furnace. In the case of $P_0 = 0$ MPa, since the cross-section of $y/D = 0.2$ is below the ejector outlet, when the particle inventory reaches a certain height in the return conduit, it can overcome the resistance of the reverse gas flow and re-enter the furnace, thus showing a higher particle concentration at $y/D = 0.2$. The higher particle concentration at $y/D = 0.2$ reflects its poor particle fluidization. It can also be seen that, at $y/D = 0.5$, the ejector operates with a higher particle concentration on the right ($x/X_0 > 0$) than on the left ($x/X_0 < 0$). However, in the cross-section at $y/D = 0.7$, the particle concentrations on the left and right sides differ only slightly. This is because the circulating particles entrained by the ejector are fully fluidized in the furnace at $y/D = 0.7$. Figure 6b shows the particle concentration distribution on Section 2. At $y/D = 0.2$ to 0.5, the particle concentrations on the left and right sides are slightly different, which is quite different from Section 1. The particle concentration in the furnace operated with the ejector has a higher and close distribution than those at $P_0 = 0$ MPa.

The above analysis shows that when the ejector is running, the particle velocity and concentration distribution on Sections 1 and 2 are basically consistent with the original distribution, and the ejector has a certain influence on their values. In the practical application of EPCG, the working medium pressure (P_p) of the ejector will affect the particle concentration and velocity and may even affect the stable particle circulation. To further analyze the relevant characteristics, and considering that the injector has a large impact on Section 1 (see Figure 5a,b), the following analysis of the effect of P_p on the gas/particle flow and pressure variation is only carried out on Section 1.

3.2 | Effect of P_p on particle velocity and concentration in return conduit

Figure 7 shows the variation of particle velocity in the return conduit under different working medium pressure P_p. As P_p increases from 0 to 0.24 MPa, the average velocity of the circulating particles in the return conduit increases from 0.95 to 2.4 m/s. It is also evident that the particle velocity in the return conduit fluctuates greatly when P_p exceeds 0.12 MPa, which indicates a decrease in flow stability.

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The change of particle concentration in the return conduit under different P_p is illustrated in Figure 8, which describes the pressure fluctuation of the experimental system from start-up to stable operation in detail. For different pressure conditions, the particle concentration shows the same trend, that is, the particle concentration first increases within 80 s, then decreases rapidly, and gradually stabilizes after 160 s. This may be due to the sharp increase in ejector entrainment with increasing P_p, thereby increasing the number of particles in the solid-phase outlet of the cyclone separator. The change in particle concentration in the return conduit tends to stabilize with time, and with the increase of P_p, the particle concentration decreases from 0.088 to 0.051 due to the increase of the circulating particles in the return conduit.

The solid circulation rate G_s is calculated using the formula $G_s = \rho_s v_s A$, where ρ_s is the particle concentration, v_s is the particle velocity, and A is the cross-sectional area of the return conduit. Figure 9 gives the G_s values at different P_p. As P_p increases, while the particle concentration decreases (Figure 8), G_s increases due to the increase in particle velocity (Figure 7). During the actual operation of EPCG, the increased G_s can increase the combustibles in the gasifier, which is beneficial to reduce the unburned combustibles of fly ash.[37] It can also be seen that the increase rate of G_s decreases significantly when P_p exceeds 0.12 MPa. This is because the particle velocity fluctuates greatly when P_p exceeds 0.12 MPa, which suppresses the increase of G_s. During the experiment, it was observed that the flow of circulating

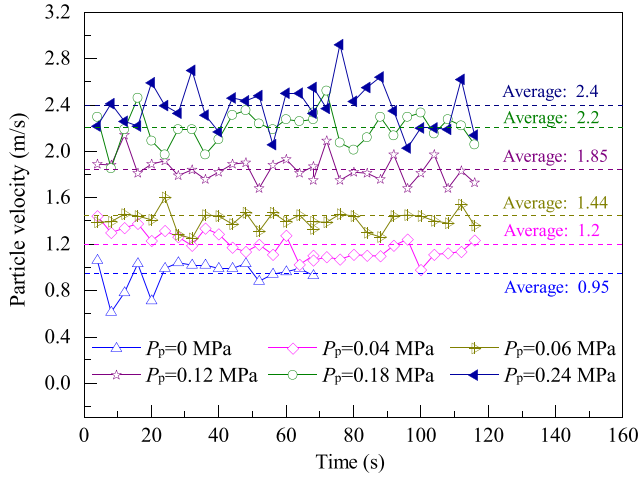


FIGURE 7 Variation of particle velocity in the return conduit under different P_p

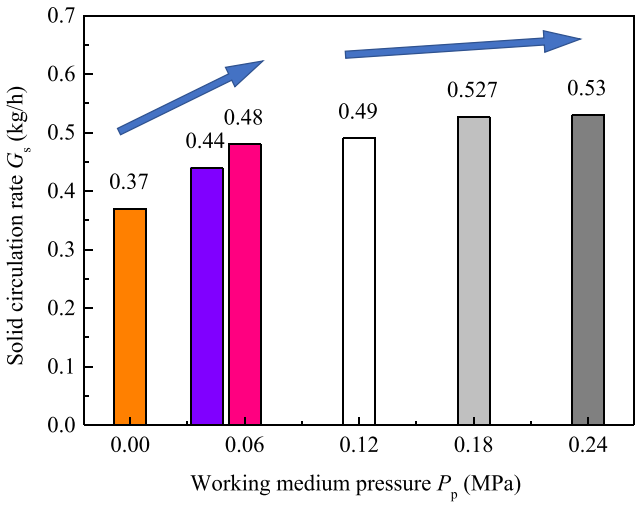


FIGURE 9 Solid circulation rate G_s under different P_p

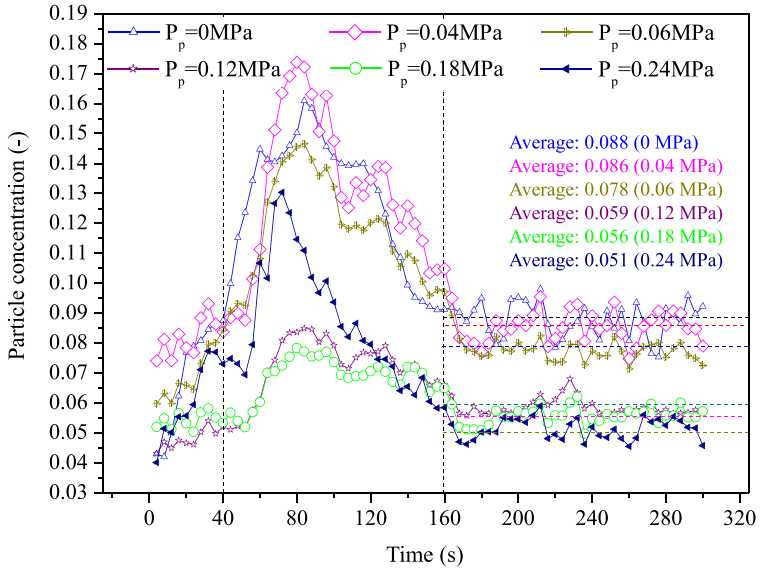


FIGURE 8 Variation of particle concentration in the return conduit under different P_p

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solids in the return conduit was intermittent, resulting in unstable solids circulation.

3.3 | Variation of particle velocity and concentration under different P_p

Figure 10 presents the particle velocity distribution at different P_p in Section 1. Figure 10a shows the entire particle velocity curve in Section 1. Increasing P_p has less effect on the overall particle velocity distribution. To further quantitatively compare the results at different P_p, the velocities in the region of $|x/X_0| < 0.2$ are magnified and shown in Figure 10b. The particle velocity decreases gradually as P_p increases. And it can also be clearly seen that in the three sections from $y/D = 0.2$ to 0.7, the particle velocity in the region of $-0.2 < x/X_0 < 0$ is smaller than that in the region of $0 < x/X_0 < 0.2$. The specific reasons are described in Section 3.1.

Figure 11 shows the variation of the central particle velocity at $x/X_0 = 0$ for different P_p. As P_p increases from 0.04 to 0.24 MPa, the particle velocities decrease continuously at the center of $y/D = 0.2$ to 0.7. The corresponding velocities at $y/D = 0.2$, 0.5, and 0.7 drop from 6.22 to 5.49 m/s, 6.08 to 5.64 m/s, and 5.99 to 5.71 m/s, respectively.

Figure 12 presents the particle concentration distribution for different P_p in Section 1. Figure 12a shows the entire particle concentration curve in Section 1. To further quantitatively compare the results under different P_p, the particle concentration in the region of $|x/X_0| < 0.2$ is magnified and shown in Figure 12b. When P_p gradually increases from 0.04 to 0.24 MPa, the particle

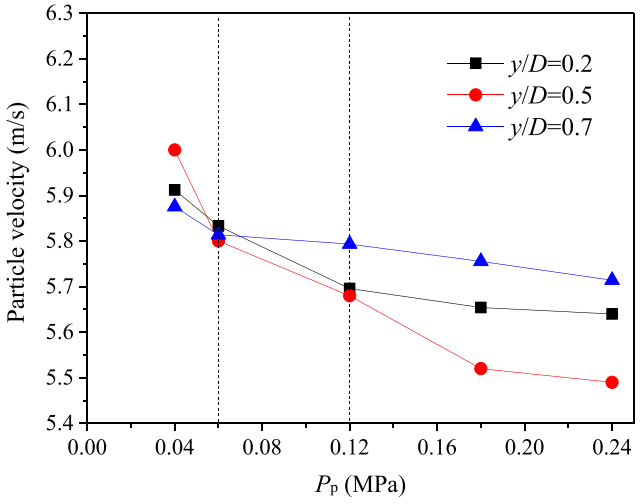


FIGURE 11 Variation of central particle velocity at different P_p

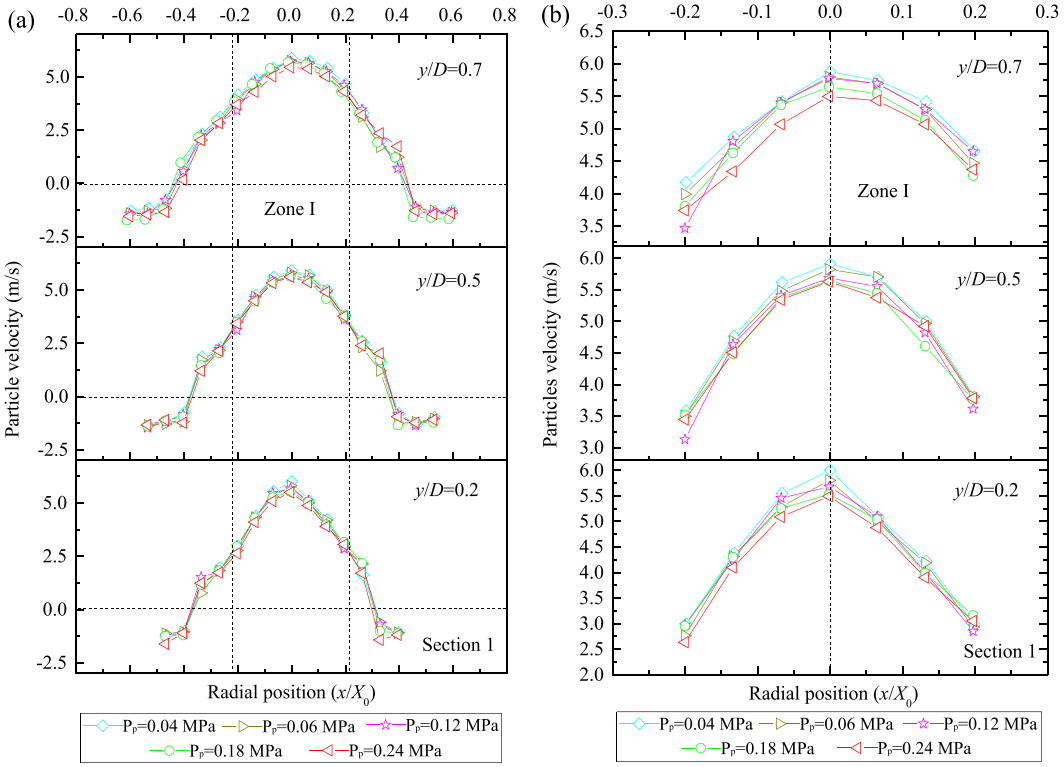


FIGURE 10 Particle velocity distribution at different P_p: (a) particle velocity curve of Section 1; (b) particle velocity curve in Zone I of Section 1

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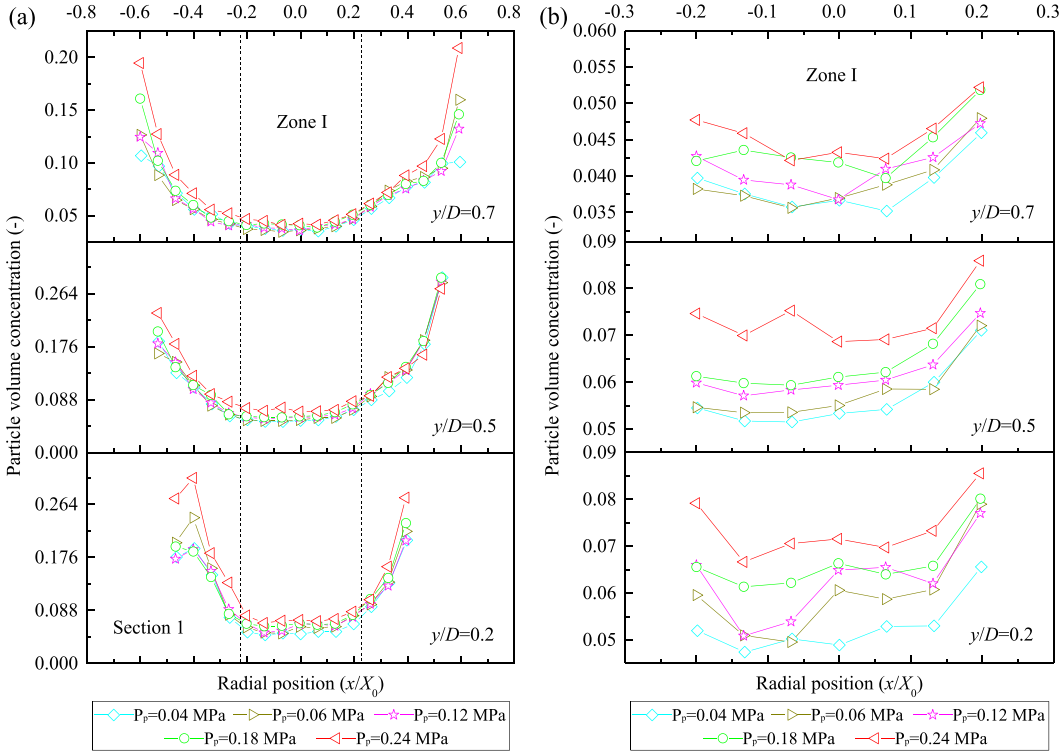


FIGURE 12 Particle concentration distribution at different P_p: (a) particle concentration curve of Section 1; (b) particle concentration curve in Zone I of Section 1

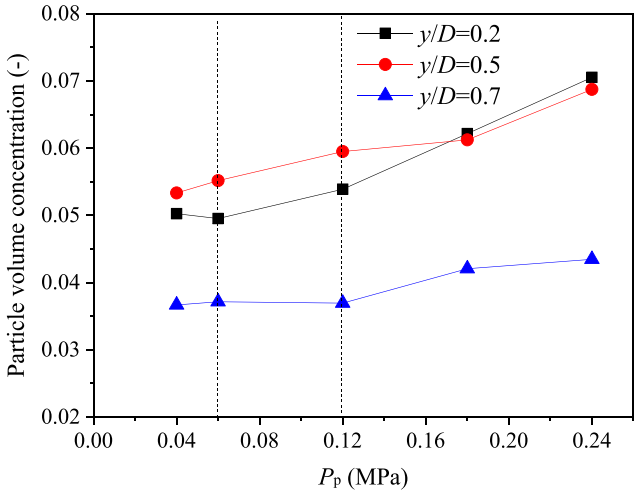


FIGURE 13 Variation of central particle concentration at different P_p

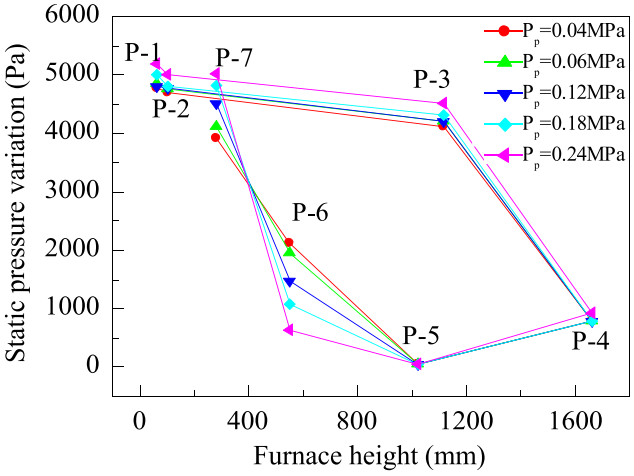


FIGURE 14 Pressure curves at different P_p

concentration in the region of $x/X_0 < -0.2$ increases significantly at $y/D = 0.2$, and the particle concentration at $x/X_0 = -0.4$ is higher than that at $x/X_0 = -0.48$. At the same time, for the section of $y/D = 0.7$, the particle concentration gradually increases, and the difference in particle concentration between the left and right sides is less than $y/D = 0.2$ and 0.5. As illustrated in Figure 12b, the particle concentration generally increases with the increase of P_p, but the particle concentration in the region of $-0.2 < x/X_0 < 0$ is less than that in the region of $0 < x/X_0 < 0.2$, reflecting the influence of the ejector on the upper gas/particle flow. Figure 13 further shows the relationship between P_p and particle concentration at $x/X_0 = 0$ on the section from $y/D = 0.2$ to 0.5. It can be seen that the particle concentration at each section increases as P_p increases from 0.04 to 0.24 MPa.

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TABLE 1 Relationships among $\triangle P$, P-7, P-2, and G_{N2} under different P_p

P_p (MPa)	0.04	0.06	0.12	0.18	0.24
$\triangle P$ (Pa)	1052	1413	2301	2991	3640
P-7 (Pa)	3920	4118	4516	4814	5012
P-2 (Pa)	4704	4753	4769	4802	4998
G_{N2} (kg/h)	2.1	2.83	4.96	7.11	9.3
G_{air} (kg/h)	309.5	309.5	309.5	309.5	309.5
G_{N2}/G_{air} (%)	0.68	0.91	1.60	2.30	3.00

The corresponding concentrations at $y/D = 0.2$, 0.5, and 0.7 increase from 0.05 to 0.07, 0.053 to 0.069, and 0.037 to 0.043, respectively.

3.4 | System pressure variation under different P_p

Figure 14 shows the pressure curves at different P_p. With the increase of P_p, the ejector outlet pressure (P-7) gradually increases and the inlet pressure (P-6) decreases, so the pressure barrier (ΔP = P-7 − P-6) provided by the ejector also increases accordingly. This is due to the increased velocity of the N_2 gas, which increases its ejection and entrainment capability, resulting in a decrease in inlet pressure.

Table 1 shows the pressure barrier ΔP, the pressures at P-7 and P-2, and the N_2 flow rate (G_{N2}) at different P_p. When P_p is 0.12 MPa, the pressure at P-7 is greater than that at P-2, and the pressure barrier provided by the ejector is sufficient to prevent backflow. According to experimental observations, particles can continuously enter the furnace from the return conduit. Table 1 also lists primary air (G_{air}) and nitrogen (G_{N2}) mass flows. According to the above analysis, the ejector has little effect on the overall flow characteristics of the particles in the furnace. Under the condition of $P_p = 0.12$ MPa, the ratio of G_{N2}/G_{air} has a small value of 1.6%, that is, a small amount of nitrogen can be used to achieve the purpose of continuous and stable pressurization.

The research results can provide important reference and guidance for the actual operation of EPCG with ejector. First, the ejector can be used to provide a stable pressure barrier and is suitable for achieving continuous solid flow in EPCG. Therefore, for the actual operation of EPCG, the installation of injector is conducive to reducing unburned combustible of fly ash and fly ash discharge and improving economic benefits. Secondly, at the beginning of the EPCG start-up, the particle concentration in the return conduit gradually stabilized after 160 s (Figure 8), which will affect the stable operation of EPCG. Special attention should be paid to the start-up phase of the ejector. In this stage, the steam flow can be gradually increased to ensure stable operation. Finally, only a small amount of working medium in the ejector is required to achieve continuous and stable pressurization. For the actual operation of EPCG, only a small amount of high-pressure steam is required, and the operating parameters of the ejector can be flexibly adjusted to optimize the EPCG performance.

4 | CONCLUSIONS

In this work, focusing on the new technology of EPCG equipped with ejector, a 1:15 gas/particle cold bench for a 45 000-Nm^3/h EPCG was established to investigate the gas/particle flow and pressure variation characteristics before and after the ejector operation and under different working medium pressures. The conclusions are as follows:

1. When the ejector operates at the design pressure $P_s = 0.187$ MPa, the outlet pressure of the return conduit is slightly larger than the bottom pressure, thereby preventing the reverse gas flow from the furnace bottom. The particle velocity and concentration distribution in the furnace are basically consistent with the distribution without the ejector, showing a similar central spout region and near-wall annulus flow region.
2. At the beginning of EPCG start-up, the particle concentration in the return conduit first increases within 80 s, then decreases rapidly, and gradually stabilizes after 160 s. As P_p increases from 0.04 to 0.24 MPa, the average particle velocity in the return conduit increases from 0.95 to 2.4 m/s, the particle concentration decreases from 0.088 to 0.051, and the solid circulation rate increases from 0.37 to 0.53 kg/h.
3. As the working fluid pressure P_p increases from 0.04 to 0.24 MPa, the ejector outlet pressure gradually increases, the inlet pressure decreases, and the

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pressure barrier provided by the ejector rises from 1052 to 3640 Pa. When P_p is 0.12 MPa, the ratio of G_{N2}/G_{air} is 0.91%, which can achieve the purpose of continuous and stable pressurization with a small amount of nitrogen.

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NOMENCLATURE

D	furnace diameter, mm
G_{air}	primary air flow rate, kg/s
G_{N2}	nitrogen flow rate, kg/s
G_s	solid circulation rate, kg/h
P_0	working medium pressure when the ejector is not running, Pa
P_p	working medium pressure when the ejector is running, Pa
P_s	design pressure of the ejector operation, Pa
ΔP	pressure barrier provided by the ejector, Pa
X_0	$D/2$, mm
x	x-axis coordinate
y	y-axis coordinate

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